Arka Mitra

AI Engineer, Swiss B Work Permit

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Overview

Technology Analyst at **Infosys** with hands-on experience delivering scalable NLP and LLM solutions for enterprise environments. Background spans optimizing production language pipelines, reducing system latency, improving test coverage, and deploying CI/CD workflows across multi-team codebases. Currently in the Infosys consulting and AI engineering onboarding track and available for immediate client engagement across financial services, insurance and enterprise automation initiatives.

Education

09.2021 – 09.2023 **M.Sc.**, *ETH Zürich*, Switzerland.
Department of Electrical Engineering and Information Technology with a specialization in Machine Learning and Signal Processing. Relevant courses include Neural Networks Theory, Probabilistic AI, AI for Good.

06.2017 – 04.2021 **B.Tech.**, *IIT Kharagpur*, India.
Department of Electronics and Electrical Communication with a Minor in Computer Science and Engineering. Relevant courses include NLP, Information Retrieval, RL, Machine Learning, Deep Learning, AI & Ethics.

Experience

11.2025 – Ongoing **Technology Analyst**, *Infosys Limited*, Zürich, Switzerland.
- Working at UBS for application refactoring on scale using agentic systems.
- Created trainings for different cohorts to get them onboarded on Generative AI.
- Preparing for client-facing AI and NLP engagements through structured Infosys onboarding.

10.2024 – 03.2025 **Senior Research Engineer**, *Telepathy Labs GmbH*, Zürich, Switzerland.
- Part of NLU team for **Stella AI**, the flagship product serving 10,000+ customers daily.
- Reduced **LLM** query latency in **Agentic** system from 3.4s to 1.2s via profiling and optimization.
- Optimized the NLU pipeline, cutting processing latency by **4x for 60%** of incoming queries.
- Migrated NLU pipeline to detect failure-prone utterances to identify urgency in human annotation.
- Integrated **multirepository CI/CD pipelines** to automate different tasks, reducing developer hours.

12.2023 – 06.2024 **Quantitative Researcher**, *Dr. Blumer and Partner Vermögensverwaltung*, Zürich, Switzerland.
- Led IT operations in the **Algosphepherd** silo, overseeing latency and data pipeline management.
- Created and deployed trading strategies for daily and intraday data, improving returns by **7.4%**
- Developed **full-stack** automated portfolio management platform for real-time analysis.

10.2021 – 09.2025 **Team Lead**, *NomadZ, ETH Zürich*, Switzerland.
- Head of vision group in **NomadZ**, a **multidisciplinary** team that trains robots to play soccer autonomously.
- Designed and deployed **end-to-end** perception pipelines with multi-sensor fusion for long term tracking.
- Built an **automated** data annotation pipeline to accelerate training data preparation across projects.
- Optimized **TensorFlowLite** inference pipelines to process 30FPS image in **23 ms**.
- Designed a human activity classifier in a **two-stage** system to provide faster inference for incoming video.

05.2021 – 09.2021 **Data and Applied Scientist**, *Microsoft*, Hyderabad, India.
- Improved the query matching accuracy by enhancing query robustness using **XLNET** instead of **BERT**.
- Integrated prices for events using **TicketMaster** API in the events tab (events near me).

04.2020 – 09.2021 **GRI MITACS Intern**, *McGill University*, Montreal, Canada.
- Used **XGBoost** and **SHAP** to identify key clinical features for COVID-19 mortality prediction.
- Worked in **cross-functional team** to deliver interpretable insights for clinicians via data-driven models.

05.2020 – 08.2021 **FT Research Intern**, *University of Turku*, Turku, Finland.
- Fused deep learning and classical ML methods to boost **small dataset** performance.
- Conducted systematic review of **XAI** techniques and ethical methodologies in medical imaging.

Relevant Projects

- **Text2SQL Conversion**
  "Converts natural language into executable SQL queries."
  - Developed an **agentic pipeline** for converting natural queries into the corresponding SQL query.
  - Achieved **64% execution accuracy** on the CORDIS dataset, beating the state-of-the-art pipeline.
  - Created a provider-agnostic LLM system that can connect to different providers like **Groq** or **Ollama**.
- **Conversational AI**
  "Agentic Framework that can perform different actions for financial advisors."
  - Designed a **full-stack** solution for financial advisors that can provide client information.
  - Combined **Agentic RAG** and different tools to provide real-time information of financial markets.
  - Developed **conversational** AI that allows communication with the system via speech.
- **Long-term Tracking of Nao Robots**
  "Multi-modal tracking of Nao robots using sensor and video data with tracklet fusion."
  - Created a hybrid tracking system using robot sensor streams and external video footage.
  - Increased robustness to noise by optimizing weight assignments for each sensor modality.
  - Fused fragmented tracks via tracklet-to-track association for life-long robot tracking.
    Advisor: **Prof. Luc Van Gool**

Skills

Languages	Python, C++, C, SQL
Libraries & Frameworks	PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, Pandas, NLTK, SHAP, FastAPI, LangChain, LangGraph, TorchServe, DVC, Azure AI
DevOps	Docker, Kubernetes, Git, CI/CD, GitHub Actions, DVC, TorchServe, AWS
Machine Learning	Natural Language Processing, Computer Vision, XGBoost, Multi-task Learning, Explainable AI, GNNs, Agentic AI, Generative AI, Ollama, vLLM
Databases	MongoDB, SQLite, PostgreSQL, Qdrant
Professional Skills	Collaboration & Cross-Functional Communication, Ownership & Reliability, Rapid Problem Diagnosis, Adaptability in High-Pressure Environments, Mentorship, Strong Work Ethic

Achievements & Academic Extra-curricular

- **2nd**, Axpo Datathon | **3rd**, GCC Competition (2024)
- Exhibitor – UN AI for Good (**'24, '25**)
- **Finalist** – HackZürich (2022)
- Grants – KIM, NCCR Automation & Robotics (2022–2023)
- MITACS Scholar (2020)
- Google NLP Summer School (2020)
- **Organizer** – **MIDL** Conference, Zürich (2022)
- **Reviewer** – **EMNLP**, FIRE, BMC (2021–2024)
- **Best UG** – Prof. Sengupta Award (2021)
- **Supervisor** – Bachelor Thesis, ETH PBL (2022)
- **TA** – D-ITET, ETH Zürich (2022–2023)

Volunteer Work

03.2020 – 12.2022 **Mentor**, *Deeplearning.ai*.
- Alpha tester; mentored learners in NLP, GANs, TensorFlow.

01.2019 – 08.2020 **Webmaster**, *IEEE, IIT Kharagpur*.
- Maintained website; co-organized tech events.

08.2017 – 06.2019 **Volunteer**, *National Service Scheme*.
- Taught village schoolchildren; Best Volunteer Award (2017).

Publications

Link for the complete list of papers: Google scholar.